POWER BI PORTFOLIO PROJECT · RETAIL ANALYTICS

RetailPulse

One interactive report for the whole retail story: sales, products, inventory health, demand forecasting and executive action.

12 report pages · 133 data visuals · 37 DAX measures used · 9 model tables

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1. Project Overview

Vision & objectives

RetailPulse is a 12-page Power BI report that turns invoice-level retail transactions into one connected view of the business: what is selling, to whom, where and when, how much stock is on hand, which products are at risk of stocking out or sitting as excess, and what demand is expected over the next 30 days. The vision is that a merchandiser, analyst or manager can open one report and move from *"what happened?"* to *"what should we do about stock?"* without writing a query.

- **Objective 1 - Visibility.** Headline KPIs (revenue, customers, products, units, invoices, average order value) on a company overview page.
- **Objective 2 - Diagnosis.** Dedicated pages for sales performance, products, categories, seasonality and customers.
- **Objective 3 - Inventory control.** Stock health, stock-out risk and overstock pages that classify every product (Stock Status, Stock Risk, Overstock Status) and quantify excess inventory value.
- **Objective 4 - Foresight.** A demand forecast page that compares actual vs forecast demand, reports forecast error (WAPE) and shows a 30-day forward forecast.
- **Objective 5 - Action.** An executive page that turns the analysis into a recommended Product Action per product.
- **Objective 6 - Self-service.** Slicers on every page (date, country, category, product, subcategory, year/quarter/month, customer) so users can explore on their own.

Target users & use cases

User	Typical question	Where in the report
Sales leadership	How are revenue, orders and average order value trending? Which countries drive revenue?	Company Overview, Sales Performance, Customer & Business Insights
Merchandising	Which products and categories earn the most? Which sell in volume but earn little?	Product Performance, Category Performance, Promotion Analysis
Inventory & supply planning	What is at risk of stocking out? What is overstocked and how much cash is tied up?	Inventory Health, Stock Risk, Overstock Analysis
Demand planners	How accurate is the forecast and what will demand look like in the next 30 days?	Demand Forecast, Seasonality Analysis
Executives	What should we act on first?	Recommendation & Executive Action
Analysts	Can I slice the numbers my own way?	Slicers on all 12 pages

Business value delivered

- **One set of numbers:** 37 measures live in the model and are reused across pages, so the same KPI reads the same everywhere.
- **Cash and service-level focus:** Excess Inventory Value and Stockout / Critical Risk counts show where money is tied up and where sales could be lost.
- **Faster decisions:** Top-N views (12 visuals) and per-product action labels point straight to the products that matter.
- **Planning horizon:** seasonality and the 30-day forecast support purchasing and stock-build decisions.

Non-functional goals

Area	Goal	How the design supports it
Accuracy	KPIs reconcile across pages and to the cleaned source	Shared measures; a single cleaned fact table (RetailPulse_Cleaned_Data)

Area	Goal	How the design supports it
Performance	Pages respond quickly to slicers	10-13 data visuals per page, Top-N filters on long product and country lists
Usability	A new user knows where to look within a minute	Same banner (title + sub-title) on every page, KPI cards across the top or side, slicers grouped together
Consistency	Twelve pages feel like one product	Fluent 2 base theme, 1920 x 1080 page canvas on all pages, shared page background colour
Maintainability	Easy to extend and audit	Descriptive measure names; report saved in the folder-based PBIR format that works well with Git

2. Key Features

Each feature corresponds to a report page. Acceptance criteria are written so they can be checked directly in the report.

ID	Feature	Description	Acceptance criteria
F-01	Company Overview	Cards for Total Revenue, Total Customers, Total Products and Total Units Sold; Top 5 products by revenue; monthly revenue trend; Top 10 countries by revenue; date, country and product slicers.	All visuals respond to the three slicers; Top-N visuals show 5 / 10 items.
F-02	Sales Performance	Cards for revenue, units, invoices and average order value; monthly sales trend; revenue by country and by product; invoice-level transaction table; period, country and category slicers.	Card totals equal the sum of the country chart for the same filters; the table drills to individual invoices.
F-03	Product Performance	Revenue per product, units, product count and invoice cards; Top 10 by revenue and by quantity; revenue-vs-quantity scatter; product detail table.	Top-N charts respect the date and country slicers; the table lists revenue, units and invoices for each product.
F-04	Category Performance	Category revenue bar and donut, category revenue share and top-category cards, monthly revenue trend by category, category comparison matrix.	Category shares add up to 100%; the matrix and donut agree on category revenue.
F-05	Inventory Health	Inventory value, on-hand and on-order cards; stock status table; days of cover and on-hand charts; inventory value and units by category.	Every product in the stock table has a Stock Status and Days of Cover; slicers filter all visuals.
F-06	Stock Risk Analysis	Stock-out, critical-risk, high-risk and healthy product counts; risk distribution donut; stock risk by category; inventory vs reorder point scatter; product risk table.	Risk counts match the number of products in the Product Risk table for the same filters.
F-07	Overstock Analysis	Excess units, excess value, inventory value and overstock product cards; overstock product table; excess value by category; inventory vs demand scatter.	Excess Inventory Value on the card equals the sum of the category chart.
F-08	Promotion Analysis	Revenue, units, transactions and average selling price cards; monthly revenue trend; Top 10 products; units by country; transaction detail table.	All visuals respond to the invoice-date, category and product slicers.
F-09	Seasonality Analysis	Revenue, units and peak-month demand cards; monthly trend with year-over-year growth; quarterly and monthly demand columns; year, quarter and month slicers.	Quarterly totals add up to the yearly total; slicers on year / quarter / month filter every chart.
F-10	Demand Forecast	Actual vs forecast demand cards; Forecast WAPE %; actual-vs-forecast line; WAPE by product; future 30-day forecast area chart; Top 10 products by future demand; back-test detail table.	The product slicer filters all forecast visuals; WAPE is shown per product and overall.
F-11	Customer & Business Insights	Unique and repeat customers, average revenue per customer, average units per invoice; purchase frequency and revenue distribution; Top 10 countries by customer revenue; monthly active customers.	Frequency and revenue-group charts add up to the unique-customer count.
F-12	Recommendation & Executive Action	Products requiring review, potential shortage, overstock and healthy counts; Product Action matrix; recommendations by action and by category; inventory risk vs daily demand scatter.	Every product has exactly one Product Action; action counts reconcile with the cards.
F-13	Consistent experience	Fluent 2 base theme, a banner shape and title text box on every page, 1920 x 1080 canvas.	Title position, banner and card style match on all 12 pages.

3. Technology Stack

Category	Technology	Rationale / alternatives
BI authoring	Power BI Desktop (project file saved in the PBIR report format, report schema 3.3.0)	Native visuals, DAX engine and a direct route to sharing. Alternatives: Tableau, Qlik Sense, Looker Studio.
Data preparation	Cleaning done before modelling (fact table is named RetailPulse_Cleaned_Data). *Tool: confirm - Power Query / Python / MySQL.*	Keeps the model free of bad rows (e.g. blank customers, negative quantities).
Data model	9 tables: transaction fact, Calendar, Customer_Summary, ProductCategoryMapping, Inventory, Product_Master and three forecast tables	Separates transactions, time, customers, stock and forecasts so measures stay simple.
Calculations	DAX measures (37 used in visuals) plus derived columns	Measures recalculate with every slicer; classification columns (Stock Risk, Overstock Status, Product Action) make results easy to filter.
Time intelligence	Calendar table (Year, Quarter, Month Name, Month Year) and a Year-over-Year Growth % measure	Correct month ordering and period comparison.
Inventory logic	Inventory table: On Hand, On Order, Reorder Point, Target Stock, Avg Daily Units Sold, Days of Cover, Unit Cost	Turns sales velocity into stock-cover and risk classification.
Forecasting	Forecast_Test_Results, Forecast_Metrics (WAPE), Future_Demand_Forecast tables loaded into the model. *Method: confirm.*	Forecast accuracy is reported with WAPE, which works well for intermittent product demand.
Visualisation	133 native visuals in 10 visual types: cards, slicers, bar, column, line, area, donut, scatter, table, matrix	Native visuals are fast and cross-filter automatically.
Design system	Fluent 2 base theme (Fluent2-CY26SU08), banner shapes and title text boxes	Consistent, clean look. Alternative: corporate theme JSON.
Publishing	Power BI Service. *Status: confirm.*	Sharing, scheduled refresh and usage metrics.

4. Architecture Overview

Data moves through seven stages, from raw transaction lines to business users. The project is a descriptive and diagnostic analytics solution with a forecast layer: the forecast tables are loaded into the model and visualised, and their accuracy is tracked with WAPE.

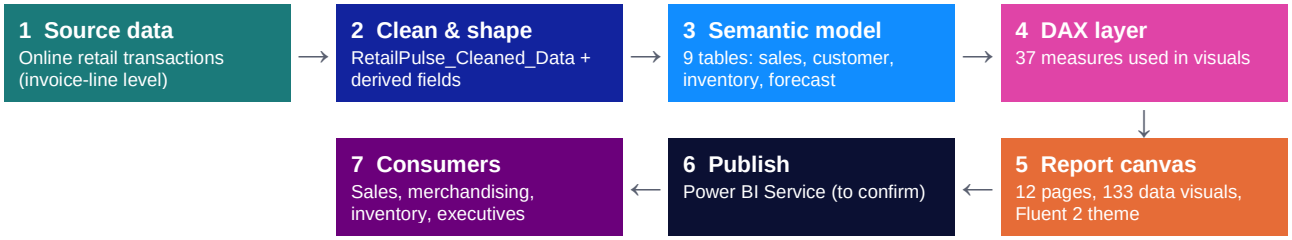


Figure 1 - End-to-end data flow.

Semantic model

The report reads from 9 tables. They fall into three groups: sales & customers (transaction fact, Calendar, Customer_Summary, ProductCategoryMapping), inventory (Inventory, Product_Master) and demand forecast (Future_Demand_Forecast, Forecast_Test_Results, Forecast_Metrics).

Sales & customers	Inventory	Demand forecast
RetailPulse_Cleaned_Data (fact)	Inventory	Future_Demand_Forecast
Calendar	Product_Master	Forecast_Test_Results
Customer_Summary		Forecast_Metrics
ProductCategoryMapping		

Figure 2 - Model tables by group. Relationship lines are not shown because relationships are stored in the compressed data model; draw them from Model view before sharing.

Tables, columns and measures used by the report

Table	Role	Columns used in visuals	Measures
RetailPulse_Cleaned_Data	Fact (transactions)	Country, Customer_ID, Description, Invoice, InvoiceDate, Price, Quantity, Revenue, StockCode, Year_Month_Label	Total Revenue, Total Units Sold, Total Invoices, Total Transactions, Total Customers, Total Products, Average Order Value, Average Selling Price, Revenue per Product, Peak Month Demand, Year-over-Year Growth %
Calendar	Date dimension	Year, Quarter, Month Name, Month Year	-
Customer_Summary	Customer grain	Customer_ID, Purchase Frequency, Customer Revenue Group	Unique Customers, Repeat Customers, Customers by Frequency, Total Customer Revenue, Average Revenue per Customer, Average Units per Invoice
ProductCategoryMapping	Category lookup	Category	Total Categories, Category Revenue Share %, Top Category Revenue
Inventory	Inventory (product grain)	StockCode, Description, Category, Subcategory, On Hand Units, On Order Units, Reorder Point, Target Stock, Unit Cost, Avg Daily Units Sold, Days of Cover, Inventory Value, Excess Units, Stock Status, Stock Risk, Overstock Status, Product Action	On-Hand Units, Total Inventory Value, Stockout Products, Critical Risk Products, High Risk Products, Healthy Products, Overstock Products, Excess Inventory Units, Excess Inventory Value, Products Requiring Review, Potential Stock Shortage
Product_Master	Product list	Product	-
Future_Demand_Forecast	Forecast (future)	Date, Product, Forecast_Demand	-
Forecast_Test_Results	Forecast (back-test)	Product, Actual_Demand, Forecast_Demand, Forecast_Error	Total Actual Demand, Total Forecast Demand, Daily Actual Demand, Daily Forecast Demand, Forecast WAPE %, Products Forecasted
Forecast_Metrics	Forecast accuracy	Product, WAPE	-

5. Execution Timeline

Add your real dates. The Power BI file does not record how long each phase took, so the 'When' column is left for you to complete. Phases and deliverables below follow what the file contains.

Phase	Major deliverables	Milestone / checkpoint	When
Discovery & data audit	Review the raw transaction file; agree the KPI list (revenue, customers, products, units, invoices, AOV, stock health); sketch page wireframes; choose theme.	M1 - KPI list and 12-page structure agreed.	[add]
Data preparation	Clean and type the transactions (RetailPulse_Cleaned_Data); create Revenue and Year_Month_Label; build Calendar, Customer_Summary and ProductCategoryMapping.	M2 - Row counts and revenue total tie out to the source.	[add]
Sales & product pages	Write base measures (Total Revenue, Units, Invoices, AOV, ASP); build Company Overview, Sales Performance, Product Performance, Category Performance.	M3 - Sales pages reviewed.	[add]
Inventory logic & pages	Build the Inventory table (On Hand, On Order, Reorder Point, Days of Cover, Stock Status, Stock Risk, Overstock Status, Product Action); build Inventory Health, Stock Risk and Overstock pages.	M4 - Every product classified; counts reconcile.	[add]
Demand, seasonality & customers	Load forecast tables and WAPE; build Seasonality, Demand Forecast, Promotion and Customer & Business Insights pages.	M5 - Forecast error (WAPE) reviewed.	[add]
Executive page, polish & publish	Recommendation & Executive Action page; Top-N filters, alignment, page names; test every slicer; publish; write documentation.	M6 - Report published; documentation signed off.	[add]

Checkpoints to apply throughout

- After modelling: tie out total revenue, invoice count and customer count to the cleaned source data.
- After each page: test every slicer, cross-filter and Top-N filter for blanks or errors.
- Before publishing: check page names, titles, alignment, colour contrast and that slicer titles match the fields they use.

6. Technical Highlights

Report build statistics

These figures are read directly from the report definition.

Metric	Value
Report pages	12 (all 1920 x 1080)
Visual containers	159 = 133 data visuals + 13 text boxes + 13 banner shapes
KPI cards / slicers	48 / 37
Charts / tables and matrices	39 / 9
Visuals with Top-N filters	12 (Top 5 on one visual, Top 10 on the rest)
DAX measures used in visuals	37 across 5 tables
Model tables / columns used in visuals	9 / 45
Visual theme	Fluent 2 base theme (Fluent2-CY26SU08)
Drillthrough	Page 1 is configured as a drillthrough target on Revenue

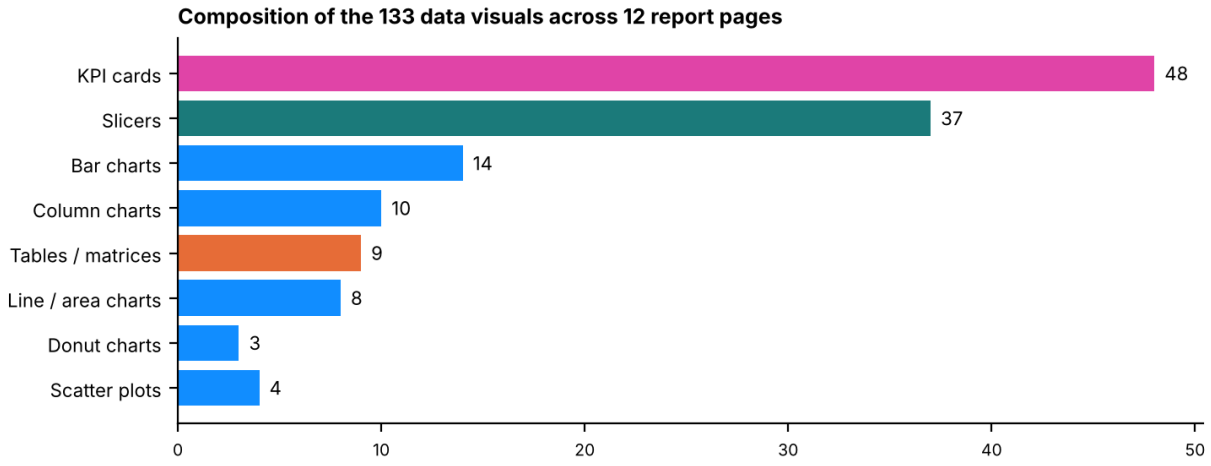


Figure 3 - What the 133 data visuals are made of.

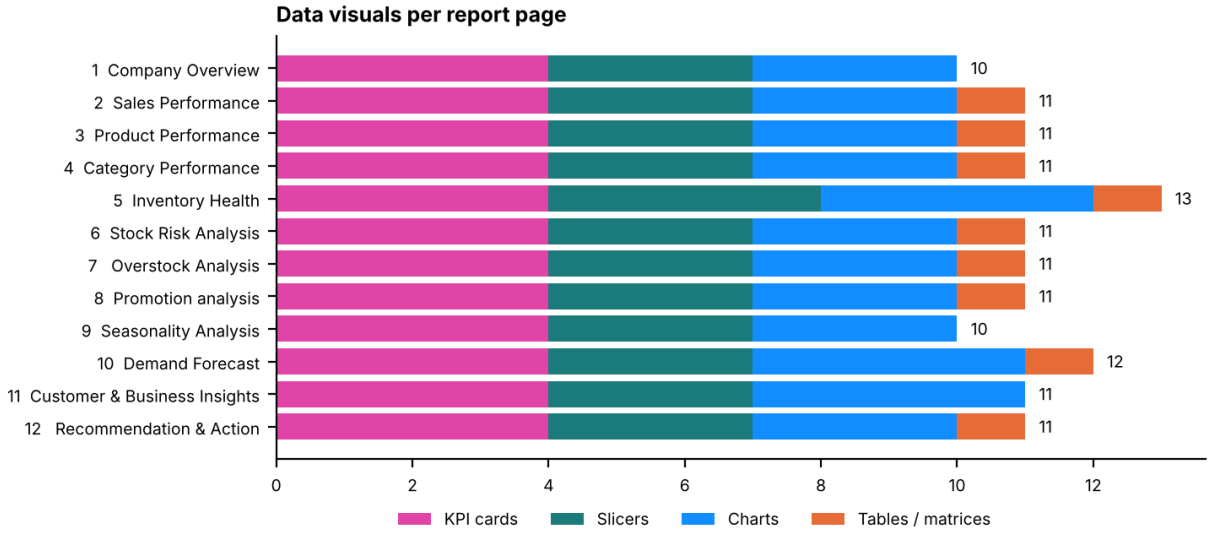


Figure 4 - Data visuals per report page.

Model metrics: forecast accuracy (WAPE)

The Demand Forecast page evaluates the forecast with **WAPE** (weighted absolute percentage error): total absolute error divided by total actual demand. It is shown as an overall card (Forecast WAPE %) and per product (Forecast_Metrics). WAPE suits retail demand because it stays stable for low-volume products, where MAPE breaks down on zero-demand days. **Add the headline WAPE value from your report card here, and the forecast method and horizon (the page shows a future 30-day forecast).**

Measure catalogue

Names are read from the report; the business question describes each measure's role. Page numbers follow the order in Appendix A (1 = Company Overview ... 12 = Recommendation & Executive Action).

Measure	Table	Business question it answers	Pages
Total Revenue	RetailPulse_Cleaned_Data	How much revenue was generated?	1, 2, 3, 4, 8, 9
Total Units Sold	RetailPulse_Cleaned_Data	How many units were sold?	1, 2, 3, 8, 9
Total Invoices	RetailPulse_Cleaned_Data	How many invoices were raised?	2, 3, 4
Total Transactions	RetailPulse_Cleaned_Data	How many transactions occurred?	8
Total Customers	RetailPulse_Cleaned_Data	How many distinct customers bought?	1
Total Products	RetailPulse_Cleaned_Data	How many products were sold?	1, 3
Average Order Value	RetailPulse_Cleaned_Data	What does a typical order bring in?	2
Average Selling Price	RetailPulse_Cleaned_Data	At what price are items actually sold?	8
Revenue per Product	RetailPulse_Cleaned_Data	How much does each product earn on average?	3
Peak Month Demand	RetailPulse_Cleaned_Data	How high is demand in the strongest month?	9
Year-over-Year Growth %	RetailPulse_Cleaned_Data	Are we growing versus the same period last year?	9
Unique Customers	Customer_Summary	How many distinct customers are in the customer table?	11
Repeat Customers	Customer_Summary	How many customers bought more than once?	11
Customers by Frequency	Customer_Summary	How are customers spread across purchase frequencies / revenue groups?	11
Total Customer Revenue	Customer_Summary	How much revenue comes from customers, by country?	11
Average Revenue per Customer	Customer_Summary	What is a customer worth on average?	11
Average Units per Invoice	Customer_Summary	How large is a typical basket?	11
Total Categories	ProductCategoryMapping	How many categories exist?	4
Category Revenue Share %	ProductCategoryMapping	What share of revenue does a category hold?	4
Top Category Revenue	ProductCategoryMapping	How much does the best category earn?	4
On-Hand Units	Inventory	How many units are in stock?	5, 6
Total Inventory Value	Inventory	How much is the stock worth?	12
Stockout Products	Inventory	How many products are out of stock?	6
Critical Risk Products	Inventory	How many products are at critical stock risk?	6
High Risk Products	Inventory	How many products are at high stock risk?	6
Healthy Products	Inventory	How many products have healthy stock?	6, 12
Overstock Products	Inventory	How many products are overstocked?	7, 12
Excess Inventory Units	Inventory	How many units are above target stock?	7
Excess Inventory Value	Inventory	How much money is tied up in excess stock?	7
Products Requiring Review	Inventory	How many products need a manual decision?	12

Measure	Table	Business question it answers	Pages
Potential Stock Shortage	Inventory	How many products may run short?	12
Total Actual Demand	Forecast_Test_Results	How much demand actually occurred in the test period?	10
Total Forecast Demand	Forecast_Test_Results	How much demand was forecast for the same period?	10
Daily Actual Demand	Forecast_Test_Results	What was actual demand day by day?	10
Daily Forecast Demand	Forecast_Test_Results	What was forecast demand day by day?	10
Forecast WAPE %	Forecast_Test_Results	How accurate is the forecast overall?	10
Products Forecasted	Forecast_Test_Results	How many products have a forecast?	10

Feature engineering

Field	Table	Purpose (inferred from name and use)
Revenue	RetailPulse_Cleaned_Data	Value of each invoice line; the base of Total Revenue and most charts.
Year_Month_Label	RetailPulse_Cleaned_Data	Month label used on monthly trend charts.
Category, Subcategory	Inventory / ProductCategoryMapping	Group products for category pages and slicers.
Avg Daily Units Sold	Inventory	Sales velocity per product; drives days of cover and reorder logic.
Days of Cover	Inventory	How many days current stock will last at the current sales rate.
Reorder Point, Target Stock	Inventory	Stock thresholds used to flag risk and excess.
Stock Status, Stock Risk	Inventory	Classify each product's stock position and the severity of stock-out risk.
Overstock Status, Excess Units	Inventory	Flag products above target stock and quantify the excess.
Inventory Value, Unit Cost	Inventory	Value of stock on hand and unit cost used to price excess stock.
Product Action	Inventory	Recommended action per product shown on the executive page.
Purchase Frequency, Customer Revenue Group	Customer_Summary	Segment customers by how often they buy and how much they spend.
Year, Quarter, Month Name, Month Year	Calendar	Time attributes for seasonality and quarterly views.
WAPE, Forecast_Error	Forecast_Metrics / Forecast_Test_Results	Forecast accuracy per product and per test row.

Field purposes are inferred from names and where they are used. Add the exact derivation logic and thresholds (for example how Stock Risk and Product Action are decided, and how On Hand and Reorder Point were produced) from Power Query / DAX.

Challenges faced & how the design handles them

Challenge	Solution in the report
Long tails of products and countries make charts unreadable	Top-N filters on 12 visuals (Top 5 / Top 10).
Stock numbers alone do not tell users what to do	Classification columns (Stock Status, Stock Risk, Overstock Status, Product Action) plus dedicated risk, overstock and action pages.
Demand is uneven across months and quarters	Seasonality page with monthly, quarterly and year-over-year views and a peak-month measure.
Forecast quality must be visible, not assumed	Back-test table, Forecast WAPE % card and WAPE by product next to the forward forecast.
Twelve pages risk looking inconsistent	Shared Fluent 2 theme, one banner and title pattern, the same canvas size and similar card layouts.
Different users need different slices	Slicers on every page (37 in total), bound to the fields each page needs.

7. Deployment & Operations

This section separates what the file shows is in place from what is recommended as the next step, so the documentation stays honest.

Area	In place	Recommended next step
Platform	Power BI Desktop project (PBIR format). Publishing status is not recorded in the file.	Publish to a dedicated workspace; share through a Power BI app for read-only users.
Refresh	Not determinable from the file	Configure scheduled refresh (daily); add a gateway if the source is on-premises.
Source control / CI-CD	None inside the file	Store the PBIR project in Git; promote Dev -> Test -> Prod with deployment pipelines.
Quality gates	Manual testing	Run Tabular Editor Best Practice Analyzer and DAX query tests before each release.
Security	Default workspace permissions	Add row-level security by country if regional managers should only see their own data.
Monitoring	None embedded	Use the Usage Metrics report; set refresh-failure alerts; track row count, blank Customer_ID share and negative quantities.

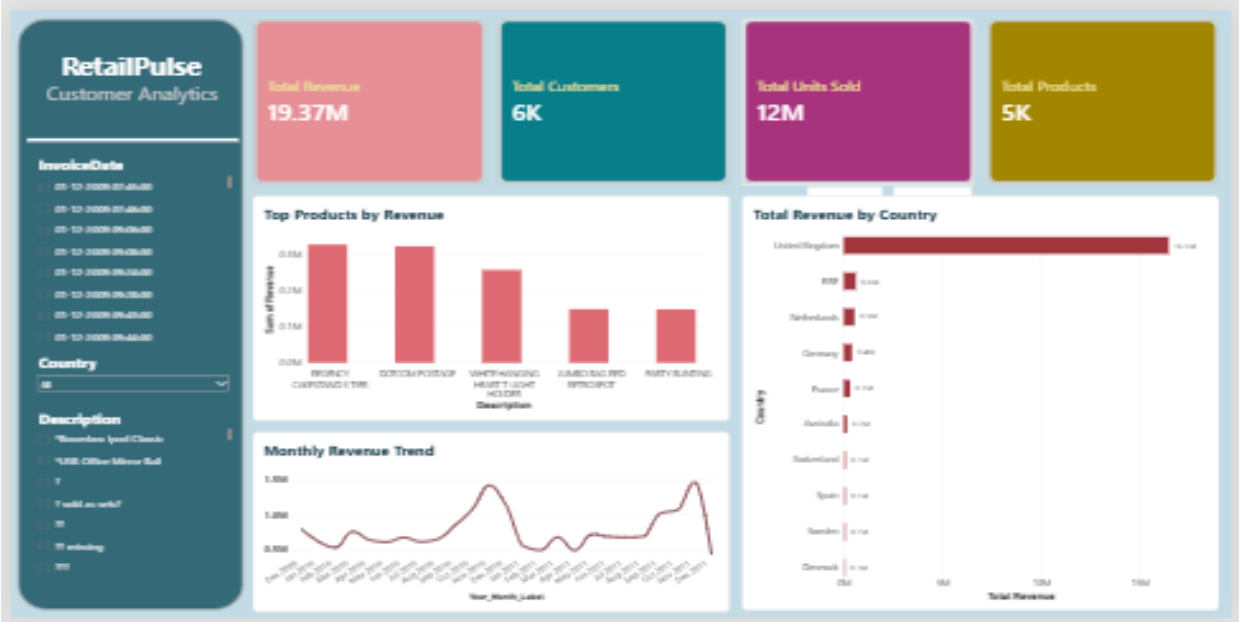
Monitoring & drift for BI

- **Data quality:** alert when row counts, blank Customer_ID rate or negative-price rows move outside normal ranges.
- **Business drift:** watch average order value, Stockout Products and Excess Inventory Value month over month; sudden shifts usually mean a data or process change.
- **Forecast drift:** track Forecast WAPE % after each refresh; a rising value means the forecast needs retraining.
- **Adoption:** review usage metrics monthly and merge pages nobody opens.

8. Visuals

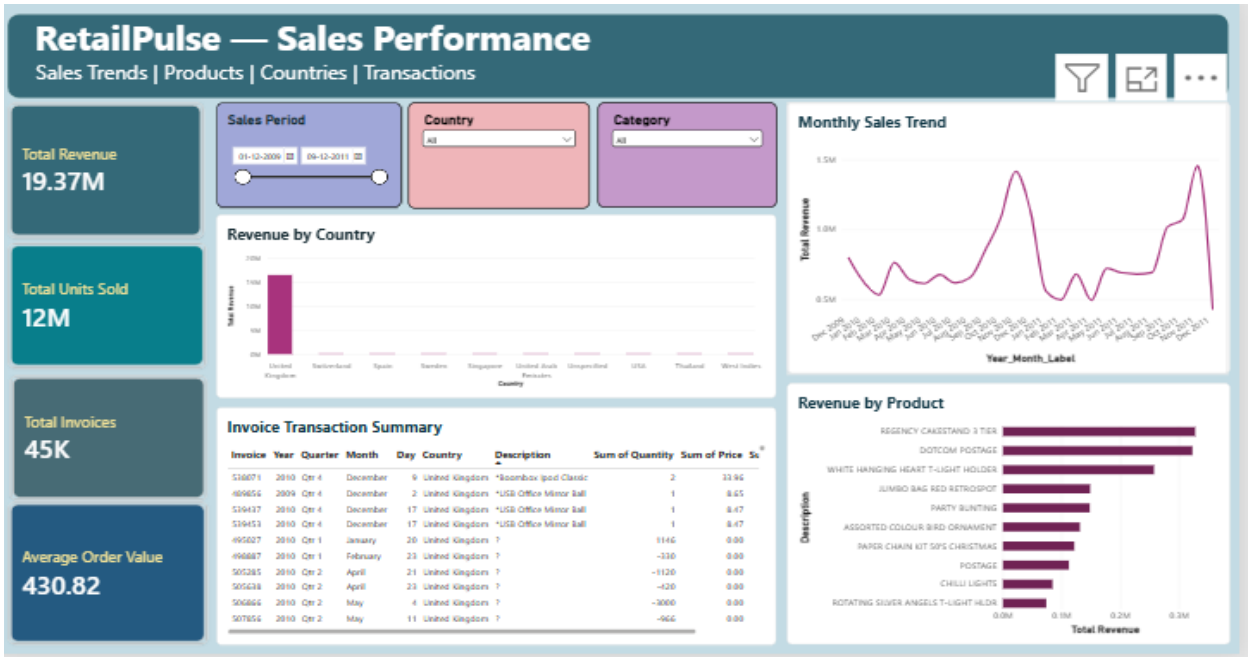
The diagrams below are layout maps generated from the real position, size and type of every visual in the report file. Replace each one with a Power BI screenshot (*File > Export > PDF*, or Windows+Shift+S on each page) for the final version.

Page 1 - Company Overview



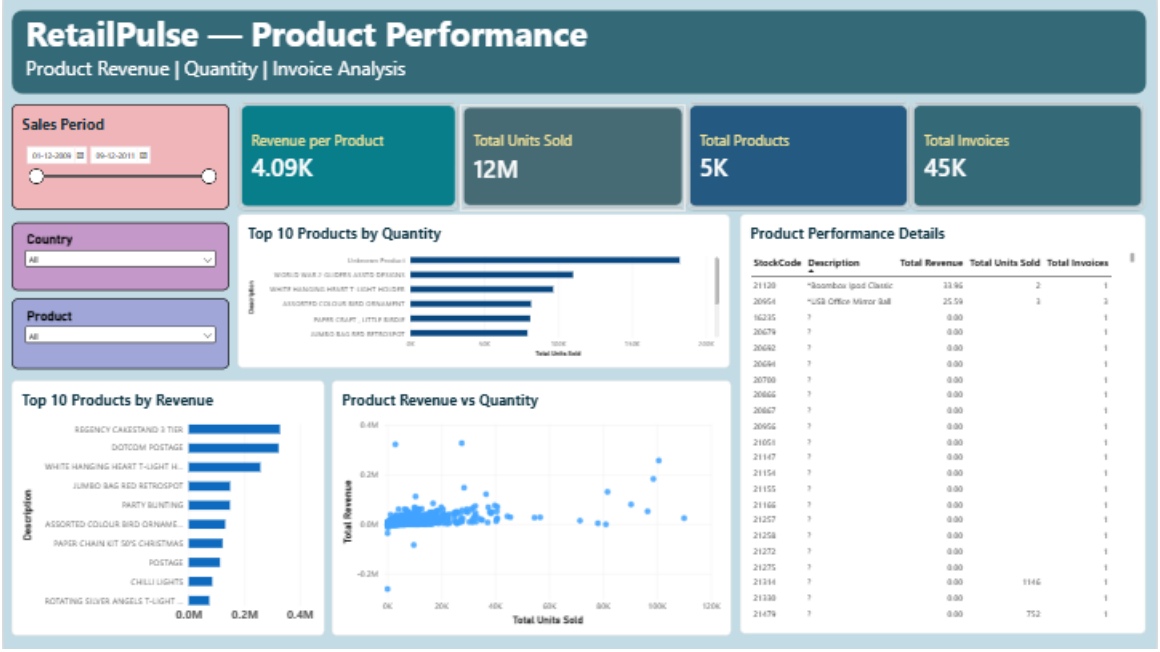
Cards: Total Revenue, Total Customers, Total Products, Total Units Sold. Top 5 products by revenue, monthly revenue trend and Top 10 countries by revenue. Slicers: invoice date, country, product.

Page 2 - Sales Performance



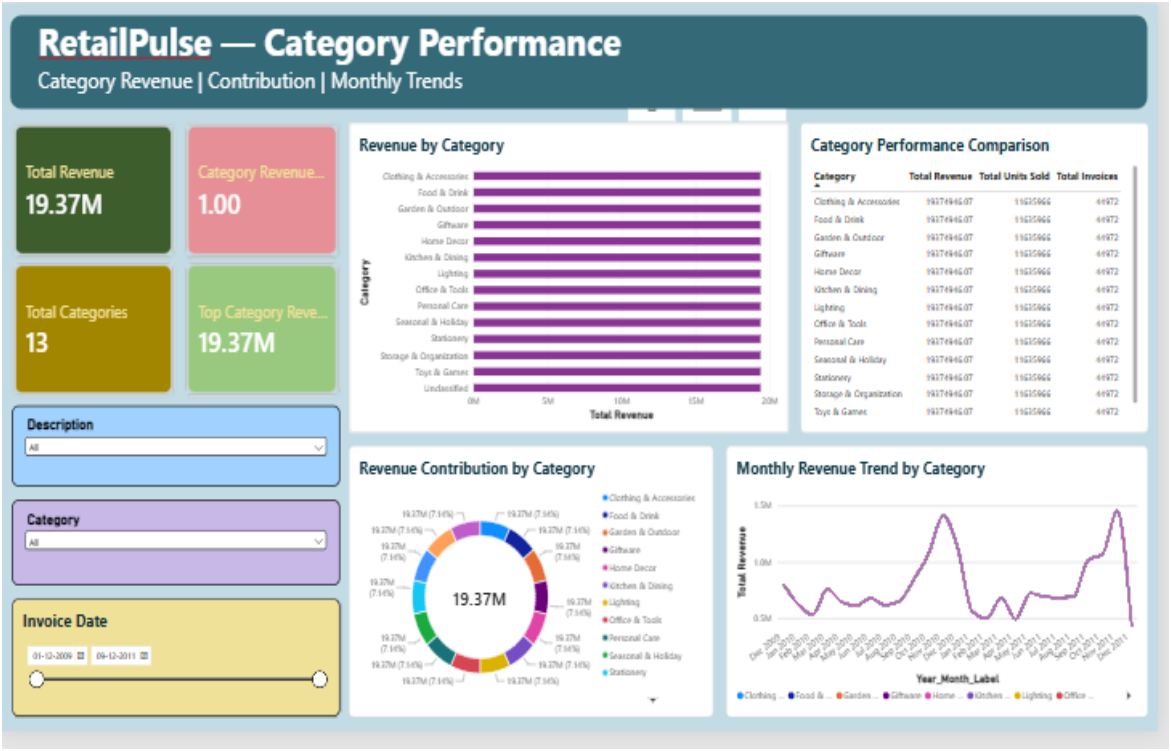
Cards: revenue, units, invoices, average order value. Monthly sales trend, revenue by country and by product, and an invoice-level table. Slicers: period, country, category.

Page 3 - Product Performance



Cards: revenue per product, units, products, invoices. Top 10 products by revenue and by quantity, revenue-vs-quantity scatter and a product detail table.

Page 4 - Category Performance



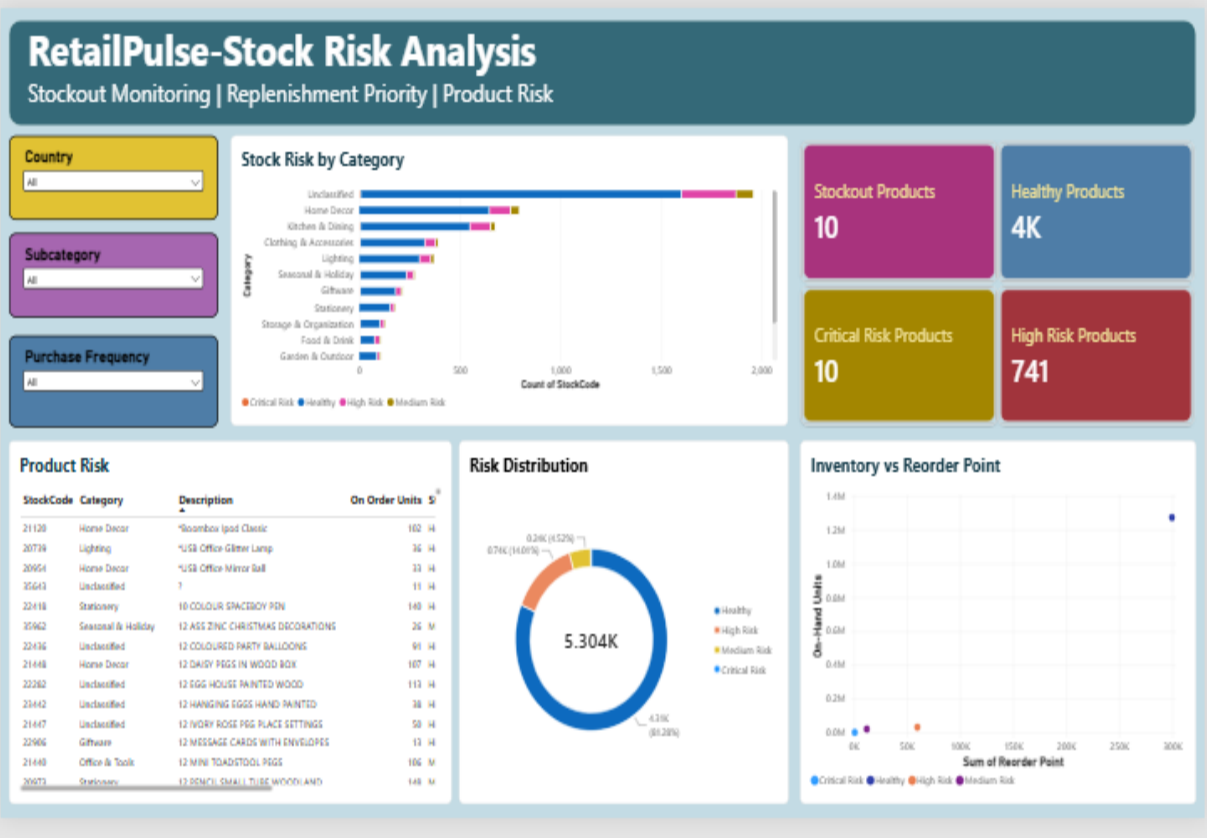
Cards: total revenue, category share, top-category revenue, total categories. Category bar and donut, monthly revenue by category and a comparison matrix.

Page 5 - Inventory Health



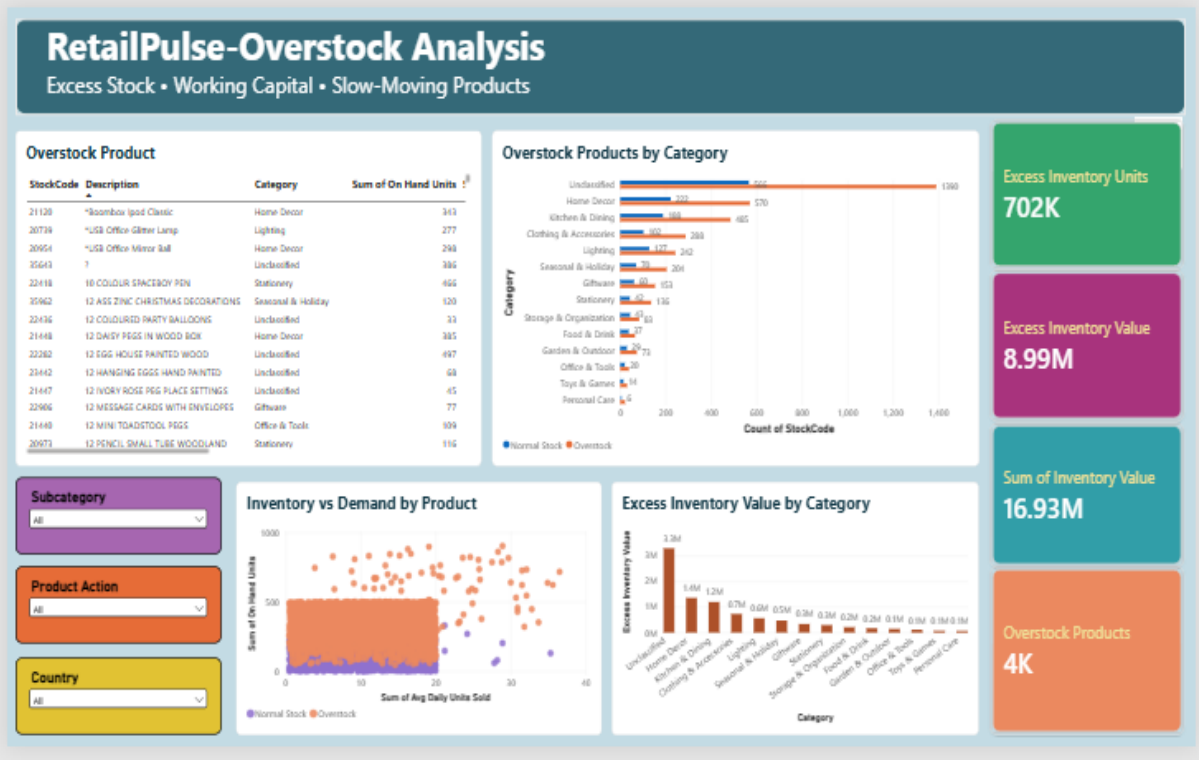
Cards: inventory value, on-hand, on-order, days of cover. Stock status table, days of cover and on-hand charts, inventory value and units by category.

Page 6 - Stock Risk Analysis



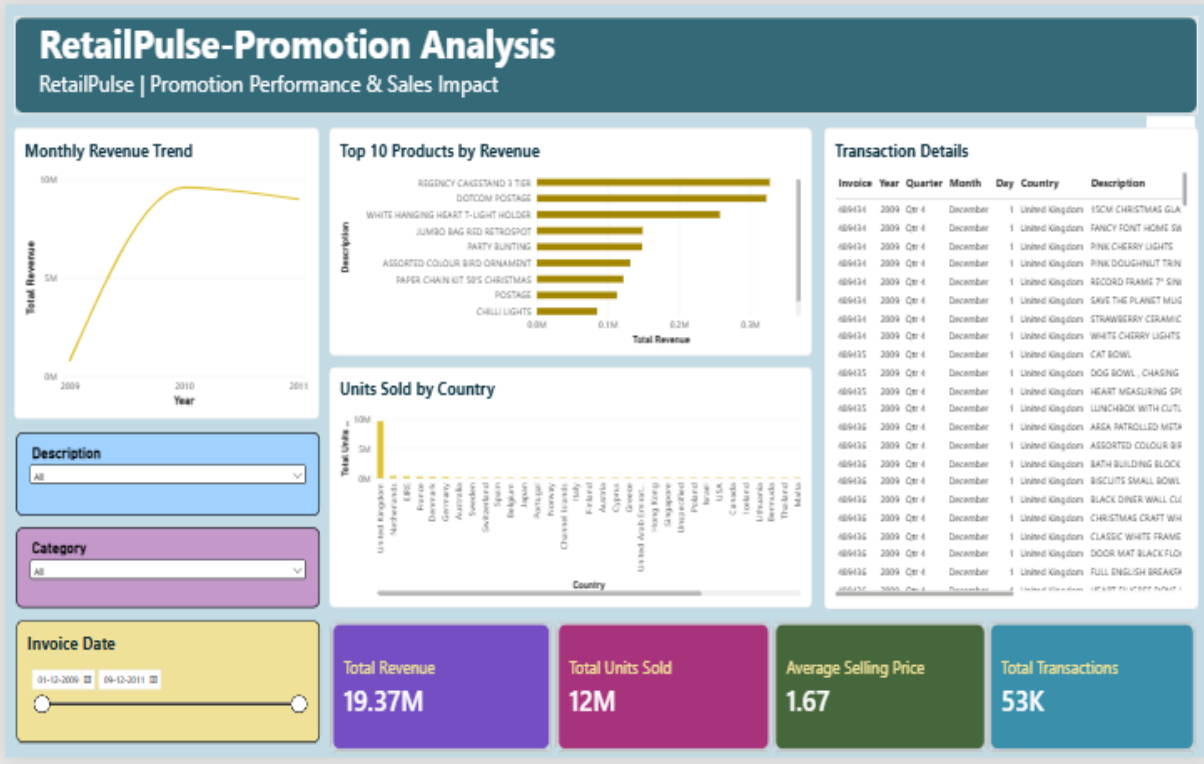
Cards: stock-out, critical, high-risk and healthy products. Risk distribution donut, stock risk by category, inventory-vs-reorder-point scatter and a product risk table.

Page 7 - Overstock Analysis



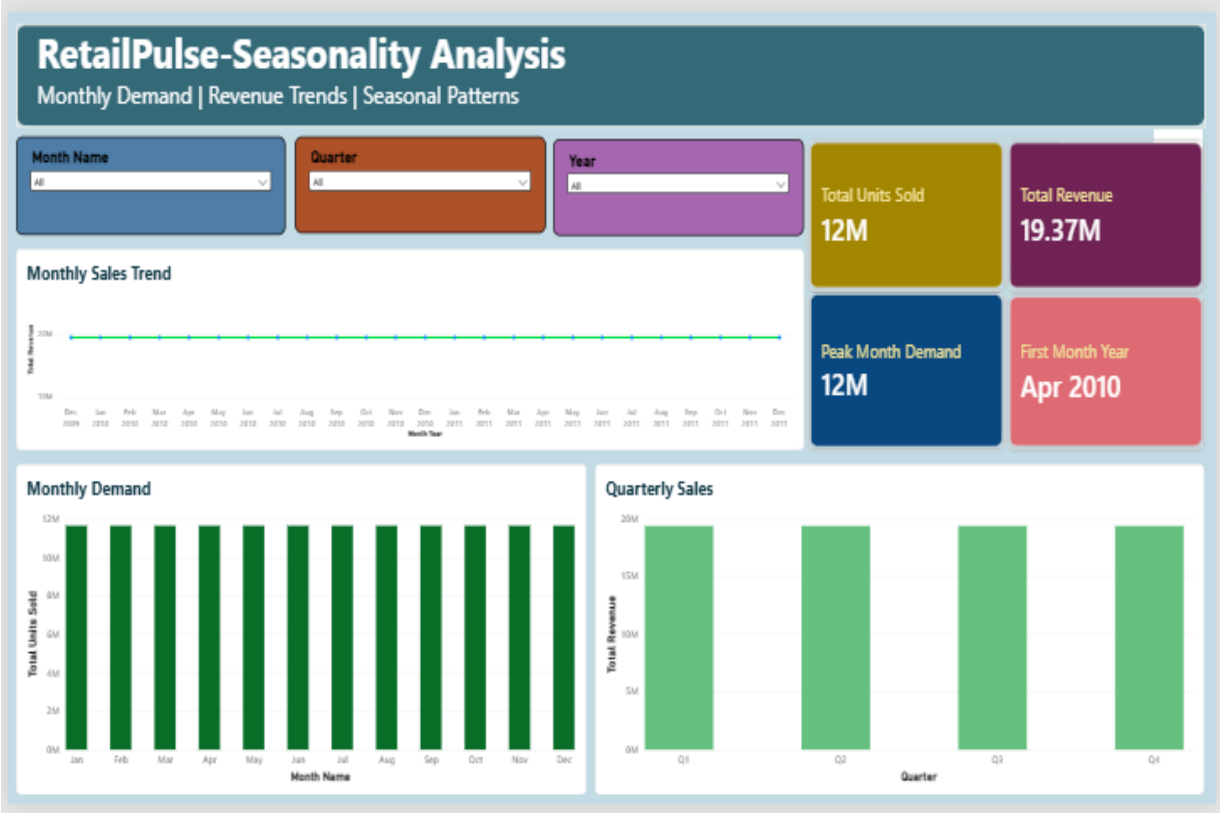
Cards: excess units, excess value, inventory value, overstock products. Overstock product table, excess value by category, inventory-vs-demand scatter.

Page 8 - Promotion analysis



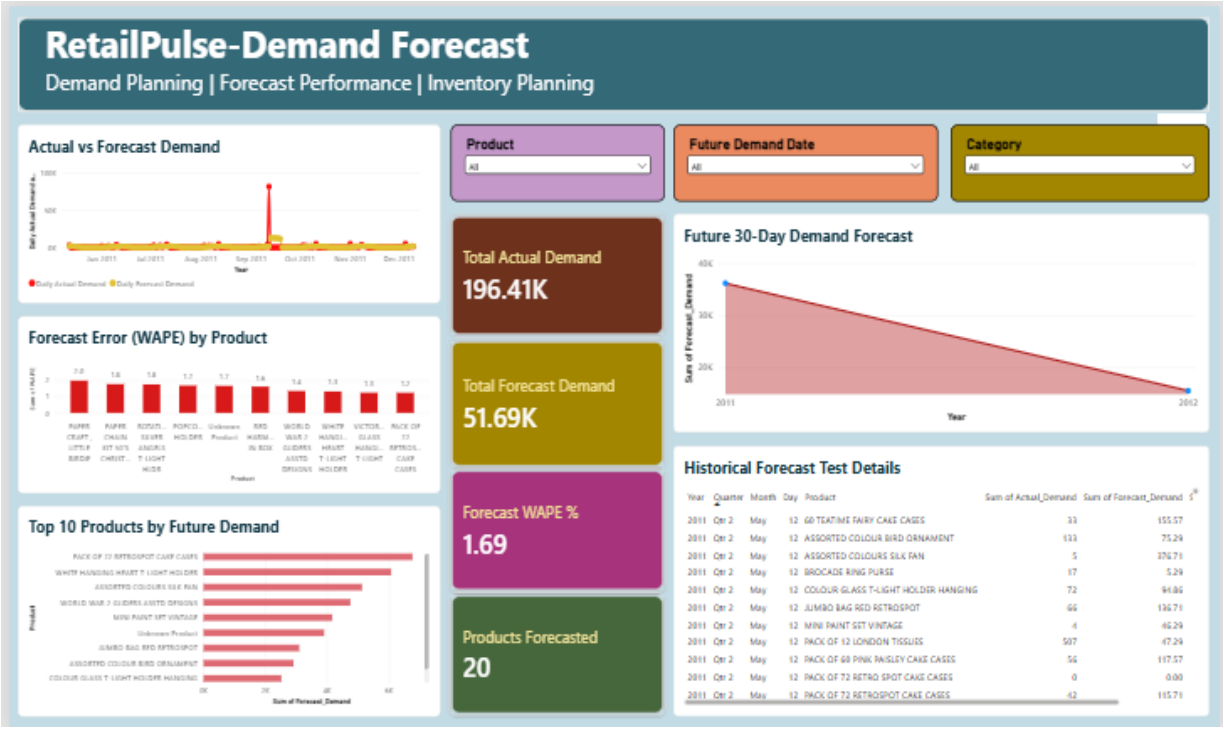
Cards: revenue, units, transactions, average selling price. Monthly revenue trend, Top 10 products, units by country and a transaction detail table.

Page 9 - Seasonality Analysis



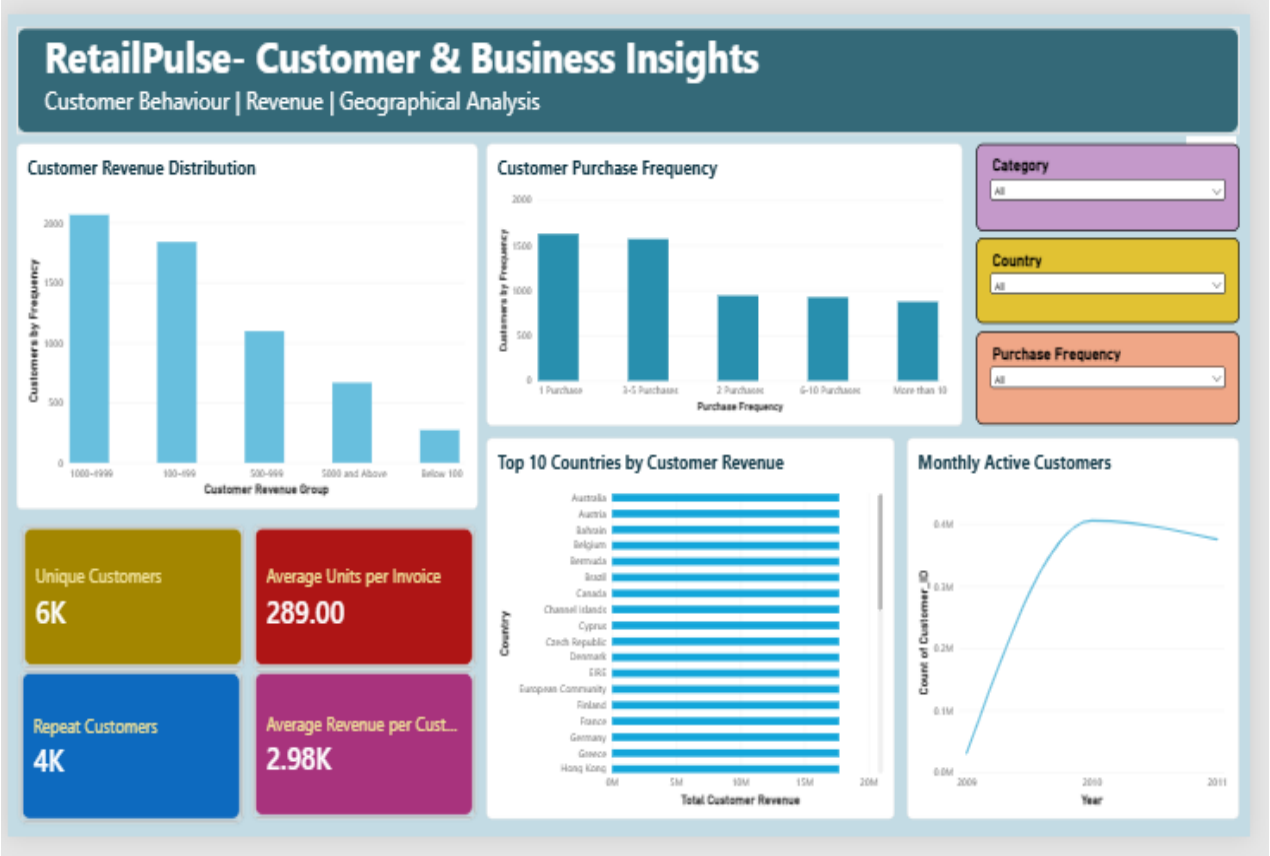
Cards: revenue, units, peak-month demand, month. Monthly trend with year-over-year growth, quarterly sales and monthly demand. Slicers: year, quarter, month.

Page 10 - Demand Forecast



Cards: actual demand, forecast demand, WAPE %, products forecasted. Actual-vs-forecast line, WAPE by product, 30-day future forecast, Top 10 future demand and a back-test table.

Page 11 - Customer & Business Insights



Cards: repeat customers, unique customers, average units per invoice, average revenue per customer. Purchase frequency, revenue distribution, Top 10 countries and monthly active customers.

Page 12 - Recommendation & Executive Action



Cards: products requiring review, potential shortage, overstock and healthy products. Product Action matrix, actions by product and category, and a risk-vs-daily-demand scatter.

9. Personal Reflection

Draft in first person. Edit freely so it reflects your own experience.

Key learnings

- A clean model makes everything downstream easier. Keeping the transaction, customer, inventory and forecast tables separate meant measures stayed short and pages stayed consistent.
- Sales data alone is not enough for inventory decisions. Adding days of cover, reorder point and risk classes turned numbers into actions.
- A forecast is only useful if its accuracy is visible. Showing WAPE next to the forecast made the limits of the model clear.
- Design is part of analytics. A shared theme, banner and card layout made twelve pages feel like one product.
- Interactivity needs testing. Slicer combinations create edge cases (blanks, empty selections) that only careful testing reveals.

Industry best practices applied

- One shared set of measures instead of ad-hoc calculations on visuals.
- A dedicated Calendar table for time and seasonality analysis.
- Top-N filters to focus attention on the products and countries that matter.
- Consistent page layout: banner title, KPI cards, slicers grouped together.
- Folder-based PBIR report format that supports source control and code review.

Future roadmap

Idea	Why it matters	Possible tools
Real promotion analysis	Add a promotion or discount flag so the Promotion page can compare promoted and non-promoted sales	Power Query, DAX
Forecast comparison	Compare forecast methods and report WAPE on a hold-out period	Python (Prophet, ARIMA), DAX
What-if inventory parameters	Let users change target days of cover or reorder rules and see the effect on risk and excess	What-if parameters, DAX
Drill-through and tooltip pages	Jump from a country, category or product to a detail page	Native drill-through, report-page tooltips
Row-level security	Regional managers see only their own markets	Power BI RLS roles
Scheduled refresh	Keep stock and sales data current	Power BI Service, gateway
Mobile layouts	Executive access on phones	Mobile layout view

Housekeeping items found in the file

These are small issues worth fixing before the report is shared. They were found by inspecting the report definition.

#	Page	Finding	Suggested fix
1	3. Product Performance	The slicer titled "Product" is bound to Future_Demand_Forecast[Date], not to a product field.	Bind it to a product column (or rename the title to "Date").
2	5. Inventory Health	The chart "On-Hand Stock vs Reorder Point" plots only On-Hand Units; Reorder Point is not in the visual.	Add Reorder Point (for example as a line or second series) or rename the chart.
3	8. Promotion analysis	No promotion-specific field is used; the page shows general sales measures.	Add a promotion / discount flag, or rename the page (e.g. "Sales Drivers").
4	1. Company Overview	The landing page is configured as a drillthrough target on Revenue, and its banner reads "Customer Analytics".	Remove the drillthrough setting if it is meant as a normal page; align the banner with the page name.
5	All pages	Page names use inconsistent numbering and spacing (for example "2. Sales Performance.", "6.Stock Risk Analysis", "12 . Recommendation...").	Rename to one pattern, such as "01 Company Overview".
6	All pages	No alt text and no data dictionary in the file.	Add alt text to visuals and a short data dictionary page or document.

Appendix A - Report Page Inventory

#	Page	Data visuals	Key fields and measures
1	Company Overview	10	Total Revenue, Total Customers, Total Products, Total Units Sold, Description, Country, Year_Month_Label, InvoiceDate
2	Sales Performance	11	Total Revenue, Total Units Sold, Total Invoices, Average Order Value, Country, Description, Invoice, Category
3	Product Performance	11	Revenue per Product, Total Units Sold, Total Products, Total Invoices, Description, StockCode
4	Category Performance	11	Total Revenue, Category Revenue Share %, Top Category Revenue, Total Categories, Category
5	Inventory Health	13	Inventory Value, On-Hand Units, On Order Units, Days of Cover, Reorder Point, Stock Status, Category
6	Stock Risk Analysis	11	Stockout / Critical Risk / High Risk / Healthy Products, Stock Risk, Reorder Point, Days of Cover
7	Overstock Analysis	11	Excess Inventory Units / Value, Overstock Products, Overstock Status, Target Stock, Unit Cost
8	Promotion Analysis	11	Total Revenue, Total Units Sold, Total Transactions, Average Selling Price, Invoice, Country
9	Seasonality Analysis	10	Total Revenue, Total Units Sold, Peak Month Demand, Year-over-Year Growth %, Year, Quarter, Month Name
10	Demand Forecast	12	Total Actual / Forecast Demand, Forecast WAPE %, Products Forecasted, WAPE, Forecast_Error
11	Customer & Business Insights	11	Repeat Customers, Unique Customers, Average Revenue per Customer, Purchase Frequency, Customer Revenue Group
12	Recommendation & Executive Action	11	Products Requiring Review, Potential Stock Shortage, Overstock Products, Healthy Products, Product Action
	Total	**133**	

Appendix B - Details to confirm before sharing

The report layer is fully documented above. The items below live in the compressed data model or outside the file, so they could not be read automatically. Filling them in will make the document complete.

Item	Where to find it in Power BI Desktop
Data source name, row count and date range	Power Query (Transform data) > source step; Data view for the fact table; Calendar min / max date.
Cleaning steps (removed rows, type changes, derived columns)	Power Query > Applied Steps for RetailPulse_Cleaned_Data.
Relationships and cardinality	Model view; add a screenshot to Figure 2.
DAX formulas for the measures	Model view / Data view > select a measure > formula bar; add key formulas (Total Revenue, Days of Cover, Stock Risk, Product Action).
How Inventory columns were produced (On Hand, On Order, Reorder Point, Target Stock, Unit Cost)	Retail sales data does not normally contain stock levels, so state whether these are derived, simulated or imported, and how.
Forecast method, horizon and headline WAPE	Python notebook or DAX behind Forecast_Test_Results / Future_Demand_Forecast; read the WAPE card on page 10.
Headline KPI values	Open the report and note Total Revenue, Customers, Products, Units on page 1.
Publishing and refresh status	Power BI Service workspace and dataset settings.
Screenshots of each page	Replace the layout maps in Section 8.